Leak Detection 1 Commented out destructor body

```
table@wkst-17:~/eece2140/lab11$ valgrind --leak-check=full ./program
==2939== Memcheck, a memory error detector
==2939== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==2939== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==2939== Command: ./program
==2939==
Original list: [3: Washington] -> [2: David] -> [1: John] -> null
After removing node 2: [3: Washington] -> [1: John] -> null
Copied list: [3: Washington] -> [1: John] -> null
Assigned list: [3: Washington] -> [1: John] -> null
==2939==
==2939== HEAP SUMMARY:
==2939==     in use at exit: 288 bytes in 6 blocks
==2939==   total heap usage: 9 allocs, 3 frees, 75,088 bytes allocated
==2939==
==2939== 96 (48 direct, 48 indirect) bytes in 1 blocks are definitely lost in loss record 4 of 6
==2939==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==2939==    by 0x10A7ED: NodeList::addNode(int, std::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >)
program)
==2939==    by 0x10A4D3: main (in /home/table/eece2140/lab11/program)
==2939==
==2939== 96 (48 direct, 48 indirect) bytes in 1 blocks are definitely lost in loss record 5 of 6
==2939==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==2939==    by 0x10AA3C: NodeList::NodeList(NodeList const&) (in /home/table/eece2140/lab11/program)
==2939==    by 0x10A55A: main (in /home/table/eece2140/lab11/program)
==2939==
==2939== 96 (48 direct, 48 indirect) bytes in 1 blocks are definitely lost in loss record 6 of 6
==2939==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==2939==    by 0x10AB53: NodeList::operator=(NodeList const&) (in /home/table/eece2140/lab11/program)
==2939==    by 0x10A59E: main (in /home/table/eece2140/lab11/program)
==2939==
==2939== LEAK SUMMARY:
==2939==    definitely lost: 144 bytes in 3 blocks
==2939==    indirectly lost: 144 bytes in 3 blocks
==2939==      possibly lost: 0 bytes in 0 blocks
==2939==    still reachable: 0 bytes in 0 blocks
==2939==         suppressed: 0 bytes in 0 blocks
==2939==
==2939== For lists of detected and suppressed errors, rerun with: -s
==2939== ERROR SUMMARY: 3 errors from 3 contexts (suppressed: 0 from 0)
```

Leak Detection 2 Deleting a node twice (double free)

```
table@wkst-17:~/eece2140/lab11$ valgrind --leak-check=full ./program
==2960== Memcheck, a memory error detector
==2960== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==2960== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==2960== Command: ./program
==2960==
Original list: [3: Washington] -> [2: David] -> [1: John] -> null
==2960== Invalid read of size 8
==2960==    at 0x49C7AE4: std::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >::~basic_string() (in /usr/l
++.so.6.0.33)
==2960==    by 0x10A757: ListNode::~ListNode() (in /home/table/eece2140/lab11/program)
==2960==    by 0x10A946: NodeList::removeNode(int) (in /home/table/eece2140/lab11/program)
==2960==    by 0x10A522: main (in /home/table/eece2140/lab11/program)
==2960==  Address 0x4e1d0f8 is 8 bytes inside a block of size 48 free'd
==2960==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==2960==    by 0x10A935: NodeList::removeNode(int) (in /home/table/eece2140/lab11/program)
==2960==    by 0x10A522: main (in /home/table/eece2140/lab11/program)
==2960==  Block was alloc'd at
==2960==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==2960==    by 0x10A81B: NodeList::addNode(int, std::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >) (in
program)
==2960==    by 0x10A481: main (in /home/table/eece2140/lab11/program)
==2960==
==2960== Invalid free() / delete / delete[] / realloc()
==2960==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==2960==    by 0x10A953: NodeList::removeNode(int) (in /home/table/eece2140/lab11/program)
==2960==    by 0x10A522: main (in /home/table/eece2140/lab11/program)
==2960==  Address 0x4e1d0f0 is 0 bytes inside a block of size 48 free'd
==2960==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==2960==    by 0x10A935: NodeList::removeNode(int) (in /home/table/eece2140/lab11/program)
==2960==    by 0x10A522: main (in /home/table/eece2140/lab11/program)
==2960==  Block was alloc'd at
==2960==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==2960==    by 0x10A81B: NodeList::addNode(int, std::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >) (in
program)
==2960==    by 0x10A481: main (in /home/table/eece2140/lab11/program)
==2960==
After removing node 2: [3: Washington] -> [1: John] -> null
Copied list: [3: Washington] -> [1: John] -> null
Assigned list: [3: Washington] -> [1: John] -> null
==2960==
```


```
==2960==
After removing node 2: [3: Washington] -> [1: John] -> null
Copied list: [3: Washington] -> [1: John] -> null
Assigned list: [3: Washington] -> [1: John] -> null
==2960==
```


```
==2960== HEAP SUMMARY:
==2960==     in use at exit: 0 bytes in 0 blocks
==2960==   total heap usage: 9 allocs, 10 frees, 75,088 bytes allocated
==2960==
==2960== All heap blocks were freed -- no leaks are possible
==2960==
==2960== For lists of detected and suppressed errors, rerun with: -s
==2960== ERROR SUMMARY: 2 errors from 2 contexts (suppressed: 0 from 0)
```

Leak Detection 3 – Accessing a node after deleting it (use after free)

```
^[[Atable@wkst-17:~/eece2140/lab11$ valgrind --leak-check=full ./program
==3005== Memcheck, a memory error detector
==3005== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==3005== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==3005== Command: ./program
==3005==
Original list: [3: Washington] -> [2: David] -> [1: John] -> null
==3005== Invalid read of size 4
==3005==    at 0x10A93E: NodeList::removeNode(int) (in /home/table/eece2140/lab11/program)
==3005==    by 0x10A522: main (in /home/table/eece2140/lab11/program)
==3005==  Address 0x4e1d0f0 is 0 bytes inside a block of size 48 free'd
==3005==    at 0x484A61D: operator delete(void*, unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==3005==    by 0x10A939: NodeList::removeNode(int) (in /home/table/eece2140/lab11/program)
==3005==    by 0x10A522: main (in /home/table/eece2140/lab11/program)
==3005==  Block was alloc'd at
==3005==    at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==3005==    by 0x10A81B: NodeList::addNode(int, std::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >) (in /home/table/eece2140/lab11/
program)
==3005==    by 0x10A481: main (in /home/table/eece2140/lab11/program)
==3005==
2
After removing node 2: [3: Washington] -> [1: John] -> null
Copied list: [3: Washington] -> [1: John] -> null
Assigned list: [3: Washington] -> [1: John] -> null
==3005==
==3005== HEAP SUMMARY:
==3005==     in use at exit: 0 bytes in 0 blocks
==3005==   total heap usage: 9 allocs, 9 frees, 75,088 bytes allocated
==3005==
==3005== All heap blocks were freed -- no leaks are possible
==3005==
==3005== For lists of detected and suppressed errors, rerun with: -s
==3005== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
```